DARPA QuEST

Robust Quantum Gadgets

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Classification of 2-local subsystem codes

Goal: Classify 2-local subsystem codes w/ eye towards use in self-correcting quantum systems

People: Gregory Crosswhite (grad, physics), Dave Bacon

Results: Developed algorithm for searching for 2-local subsystem codes and searched for codes and performed complete search for <8 qubits, partial search for >8 qubits, and searched regular lattices

- Code is polynomial in # qubits and exponential in distance of code
- Only one 4-qubit code (detecting). No error correcting codes smaller than the 9 qubit BS code. Results tell us self-correcting systems w 2-local interactions must have >=9 qubits.

Paper in preparation describing algorithm and results
(expected finish Dec 09/ Jan 10)

Classical Analogy of Universal Adiabatic QC

Goal: To make FT adiabatic QC, first investigate classical analog of adiabatic QC

People: Isaac Crosson (grad, physics), Dave Bacon

Results: Defined classical model for ground state computing at finite temperature. Showed that it is not fault-tolerant at finite temperature. Showed how to make it fault-tolerant.

- Major insight: generalization of transfer matrix allows us to map thermal ensemble to classical probabilistic circuit. This allows for fault-tolerance analysis + gives efficient classical algorithms for a new class of classical spin systems

- This method can be applied to fault-tolerance in adiabatic QC

Paper in preparation (expected finish Jan 10)

Perturbation Theory Gadgets

Goal: Numeric and Analytic investigation of amplifying perturbation gadgets

People: Lukas Svec (grad, physics), Jijang Yan (grad, physics), Gregory Crosswhite (grad, physics) Dave Bacon

Results: Amplification works to amplify the energy gap, but the overlap of the system w/ the desired ground state shrinks exponentially.

• If one is to use gadget amplification to build strong interactions, one must include the ancillas in the encoding. We are currently investigating this later idea.

Internal notes made describing this, and looking for either a no-go theorem for using ancillas in the encoding or a way to get the later idea to work

Fault-tolerant Techniques for Stabilizer Hamiltonian Simulation

Goal: Fault-tolerant bang-bang simulation of stabilizer Hamiltonians

People: Dave Bacon

Results: Constructed circuits which are fault-tolerant for simulating the stabilizer Hamiltonians and verified their fault-tolerance. Next step is to analyze error model.

Notes need to be converted to a paper, deciding whether to wait for analysis of the error model

Work arising during project but not in original tasks

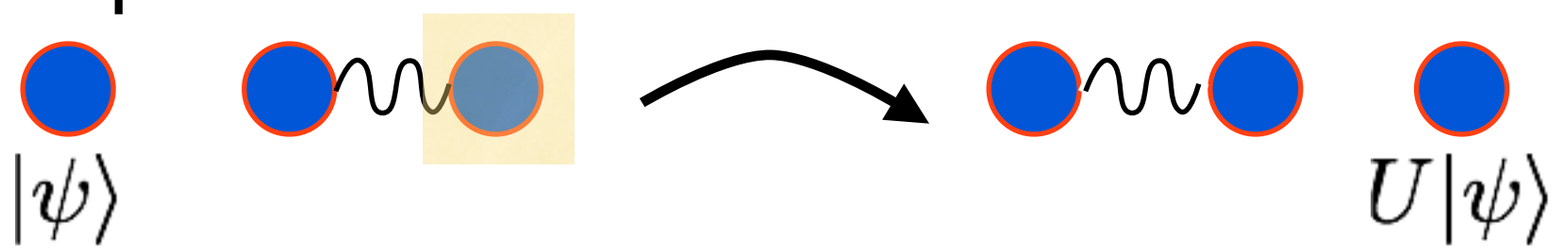
Adiabatic Gate Teleportation

People: Dave Bacon, Steve Flammia (postdoc at Perimeter Institute)

Question: how does one compute on self-correcting systems?

Idea: Make rigorous manner in which topological QC works, adiabatic evolutions

Further idea: What if you adiabatically drag a system whose energy levels are QEC codes to a system with energy level from another QEC? This spawned....



Result: brand new protocol for building a quantum computing using adiabatic evolutions! Experiments easy in dots, superconducting qubits

Paper: Phys. Rev. Lett. 103, 120504 (2009) Talk: TQC 2009, Zurich

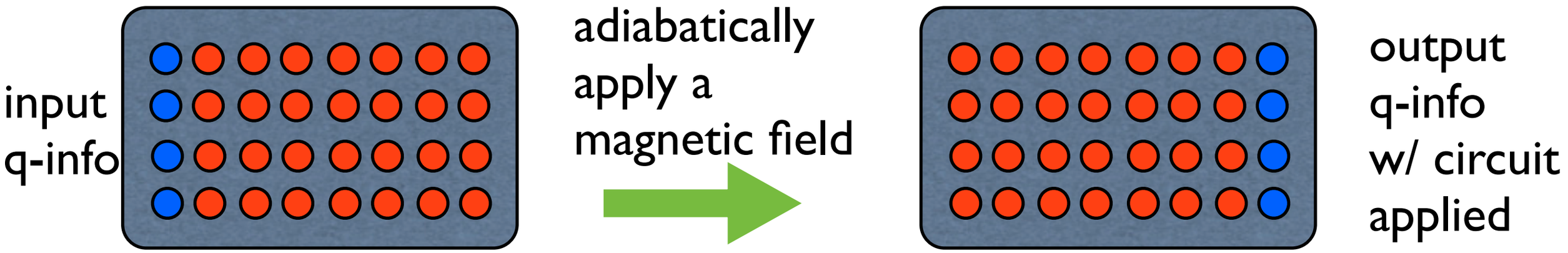
Work arising during project but not in original tasks

Quantum Transistors

People: Dave Bacon, Steve Flammia (postdoc at Perimeter Institute),

Result: We have shown how one can perform measurement based QC (a.k.a. one way QC or cluster state QC) using NO measurements, but only adiabatic evolution!
(Paper written waiting for UW Tech transfer before posting on arXiv)

Bigger result: How to build a quantum transistor!



interacting many-body quantum system

Entirely new way to build a QC: eliminates timing errors. **Would save D-wave?**

Writing paper for Nature: "How to build a quantum transistor."

Logistics / Personnel

Failed to hire postdoc last year, but have three candidates this year (Gerardo Paz Silva, Steve Flammia, Bilal Shaw)

Question: possible to run funding such that if hired in summer 2010, can fund for two years?

Adiabatic ideas (quantum transistor, gate teleportation) are looking extremely promising.

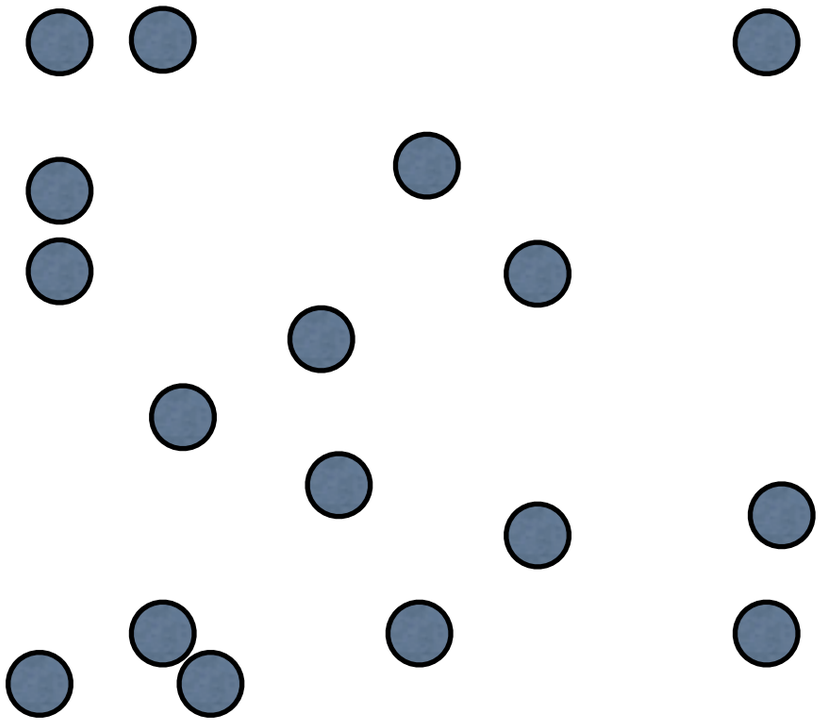
Question: add to year 2 tasks.

Extra Slides

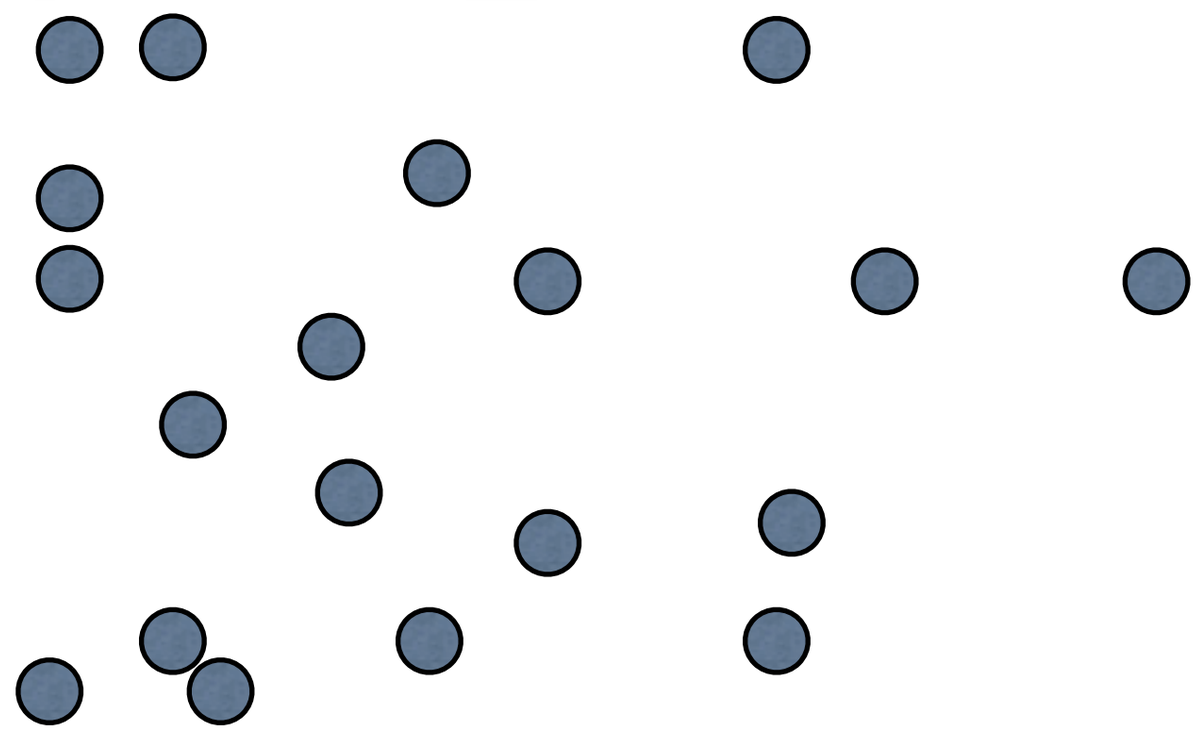
Code search


codes
4 qubits
1 code found
2b3c09d6-be64-4b5a-8c5c-55f85c35101a
5 qubits
1 code found
90553b9e-7c7a-4fb2-b6e7-d93626864e94
6 qubits
graphs (25 interesting in total)
none have vertices with degree > 4
most have degrees between 2 and 4
only a single graph with code has minbound of 3 instead of 2; it also has maxbound of 3
there is only one graph where all vertices have degree 2
EEh_
it is planar
ba27c19b-64c1-4760-bbd3-10c291524090
codes (222 interesting in total)
221 codes found with single logical
1 code found with two logicals
ba27c19b-64c1-4760-bbd3-10c291524090
both have weight 2!!!
all vertices in graph have degree 2
this is the unique graph of 6 vertices with this property
7 qubits
graphs (216 intersting in total)
coverage
469 of 507 graphs scanned
the remaining 38 have more than 2^31 possible labelings
maybe they are redundant?
all planar graphs scanned
196 graphs with one logical
161 live on planar graphs
20 graphs with two logicals
all live on planar graphs
single graph has two weight 2 qubits
FCpd_
has 4 codes total, all of which have two qubits but only one of which has two weight 2 qubits
codes (13759 interesting in total)
13729 codes with one logical
30 codes with two logicals
in only one code do both have weight 2
3ce0d337-d14c-4184-8c28-fc048c60b4dd

Fault-Tolerant Classical GS Computing



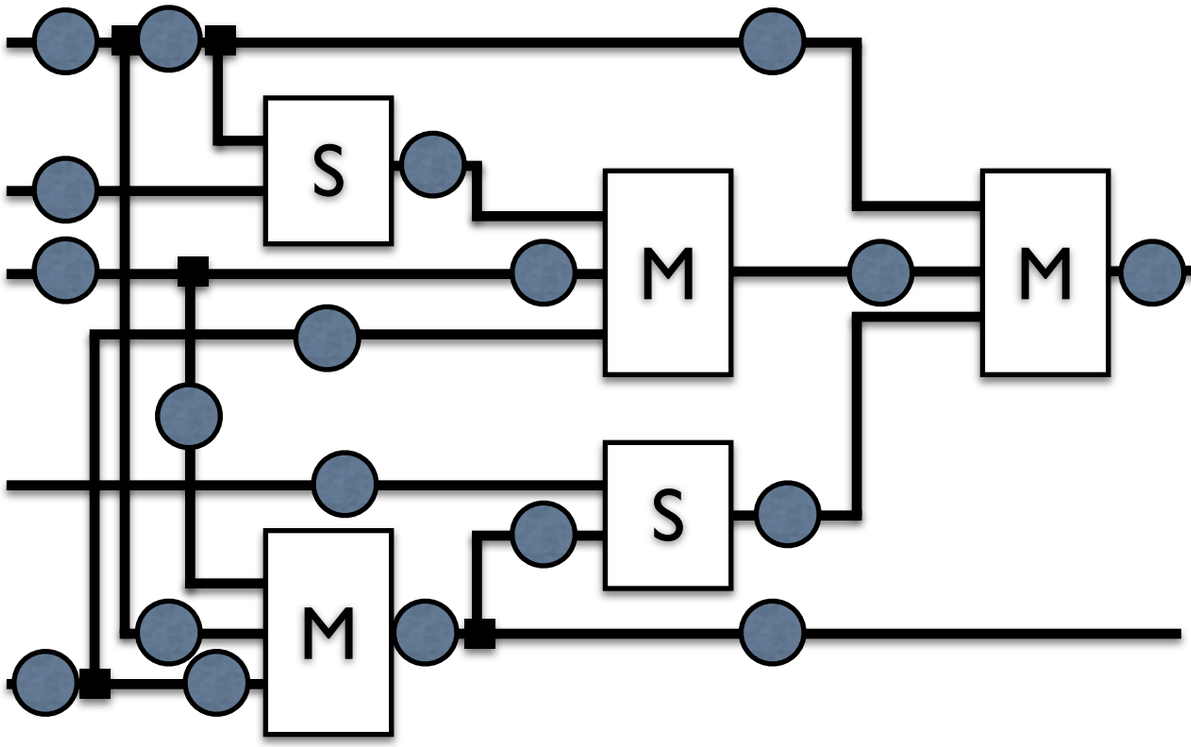
Classical circuit



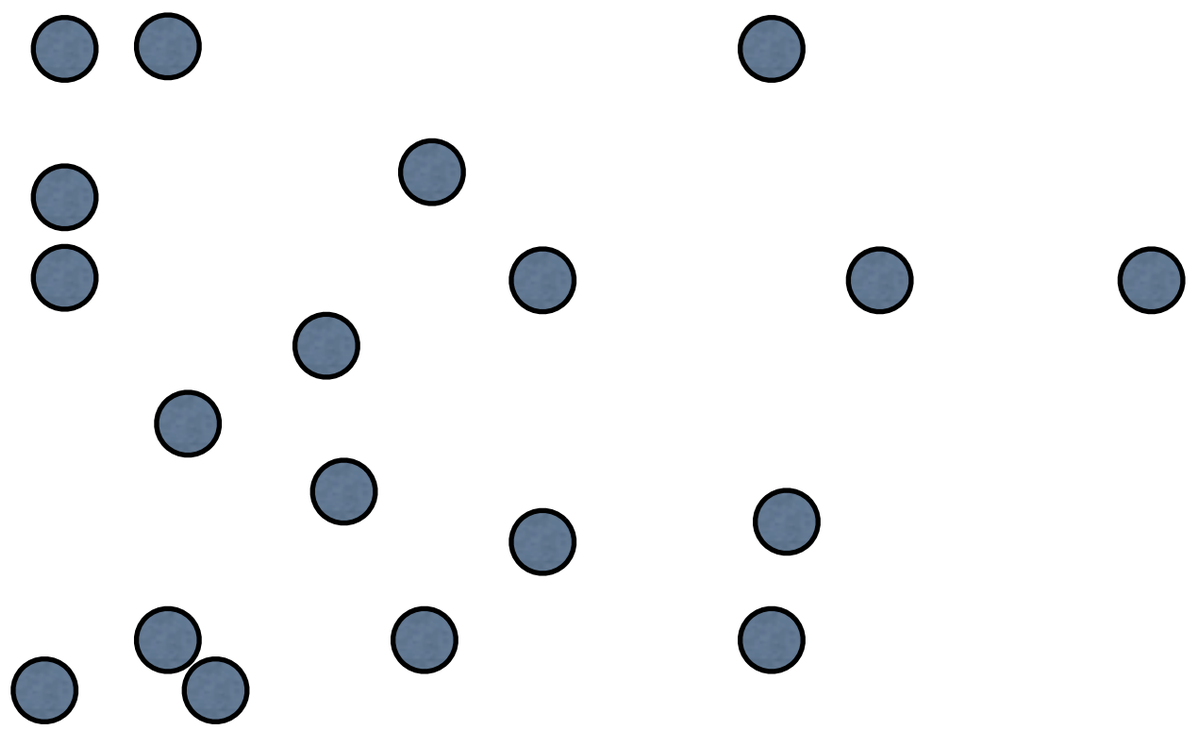
Classical spin system whose ground state = computation

But at finite temperature the circuit will fail

Fault-Tolerant Classical GS Computing



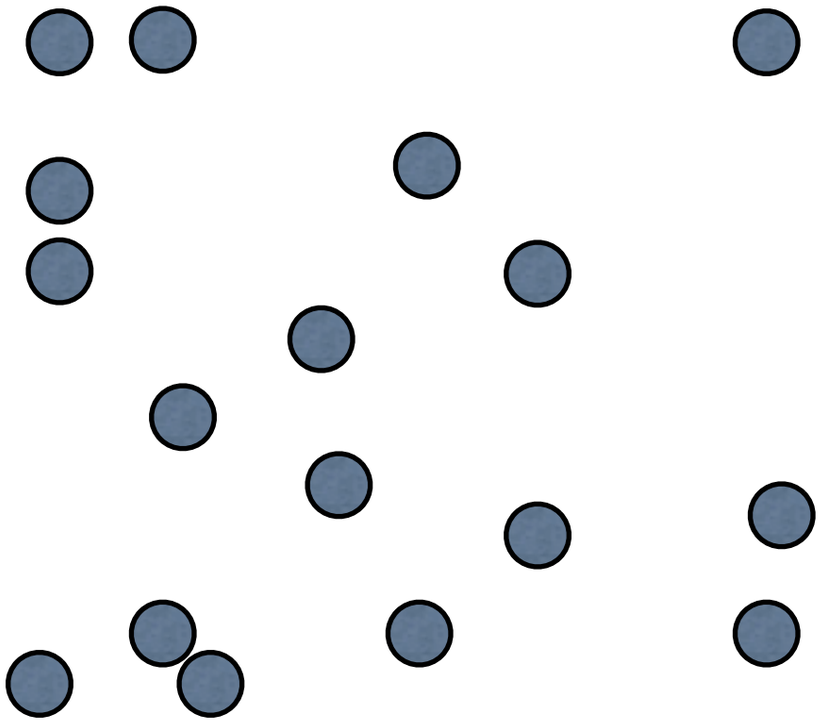
Classical circuit



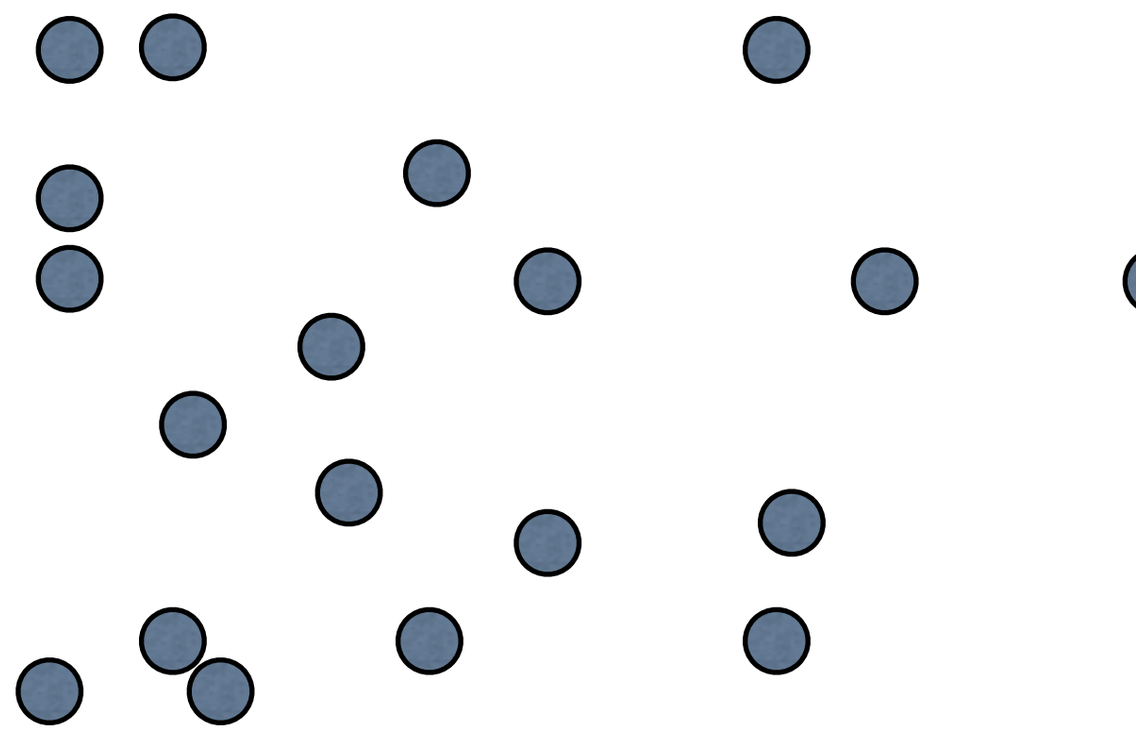
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Fault-Tolerant Classical GS Computing



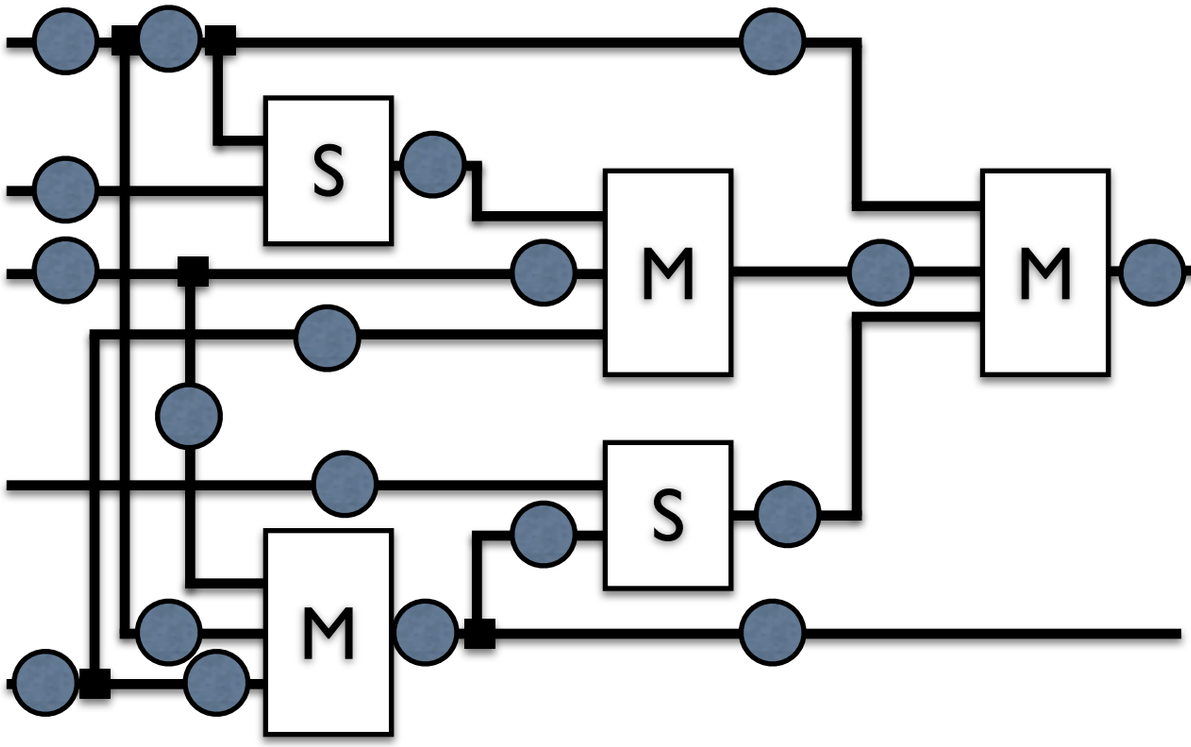
Classical circuit



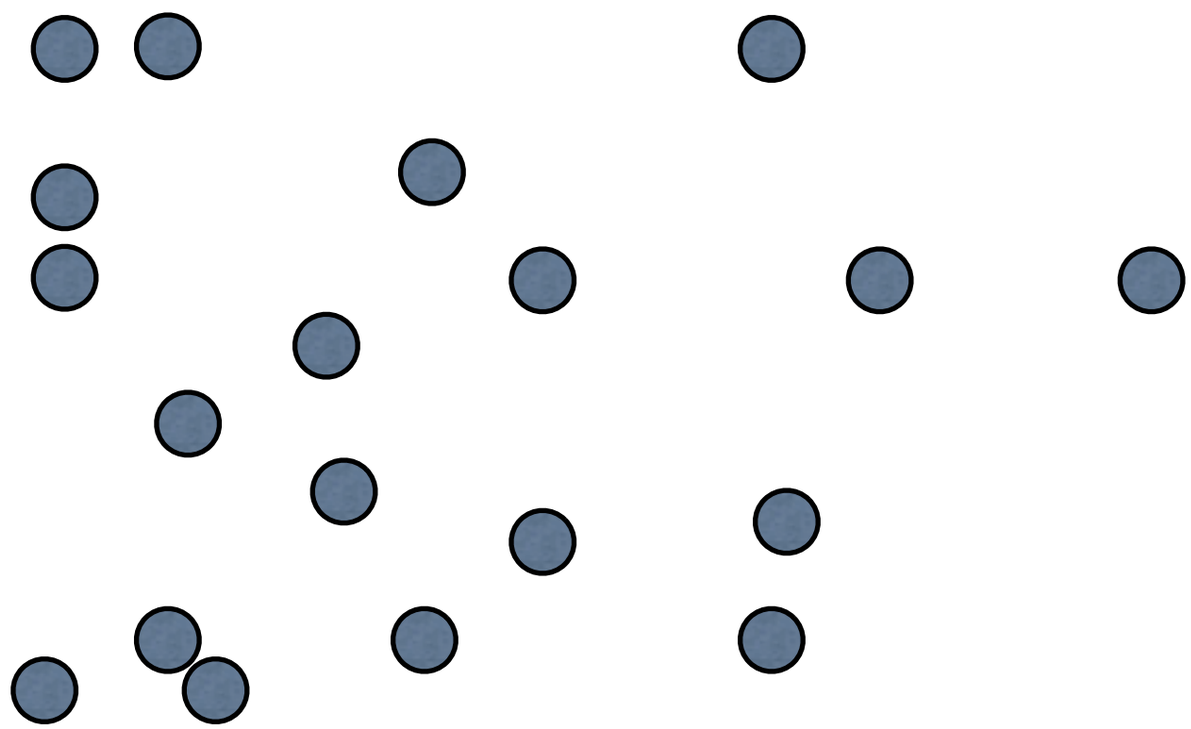
Classical spin system whose ground state = computation

Insight: at finite T, partition function of system on right is exactly related to circuit on left with probability of failure related to temperature.

Fault-Tolerant Classical GS Computing



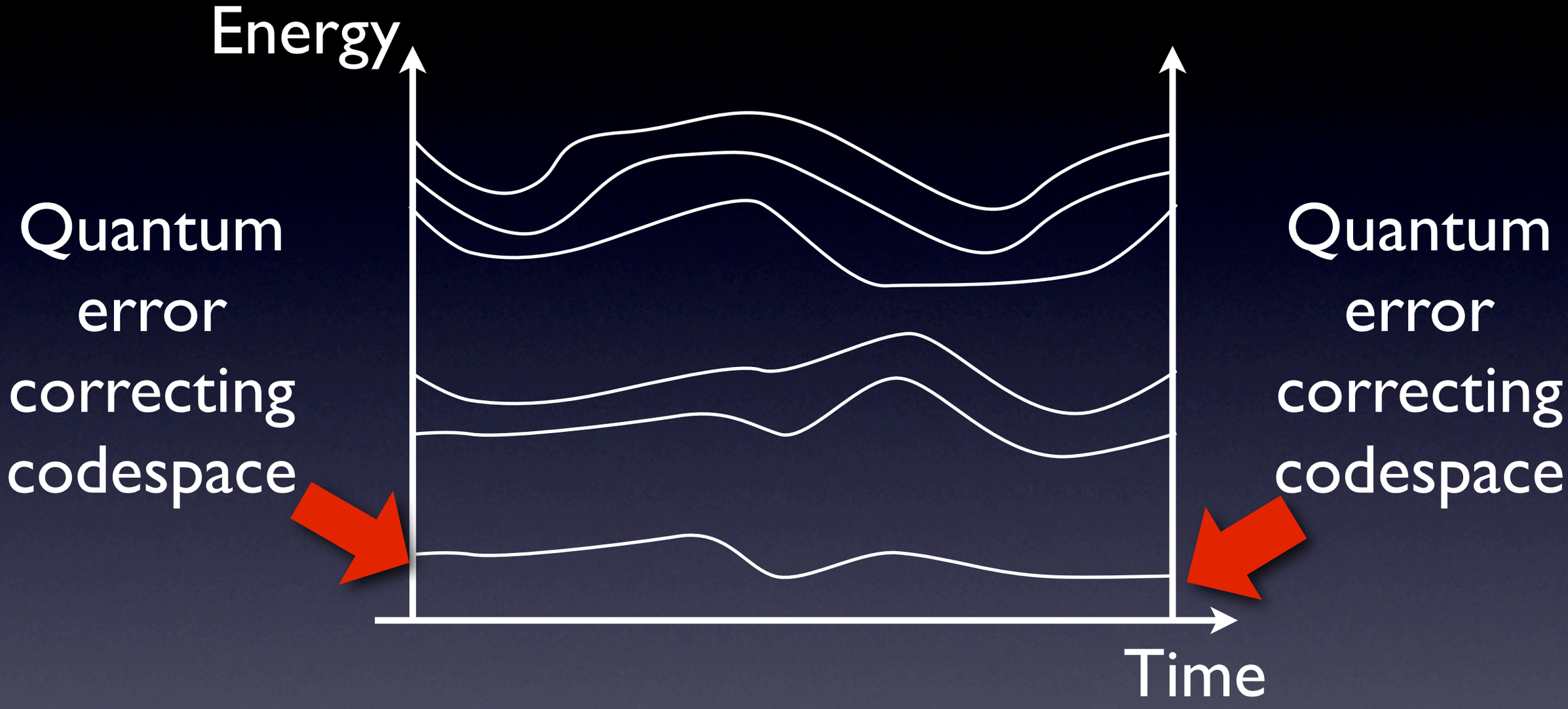
Classical circuit



Classical spin system whose ground state = computation

Insight: at finite T, partition function of system on right is exactly related to circuit on left with probability of failure related to temperature.

Adiabatic Dragging

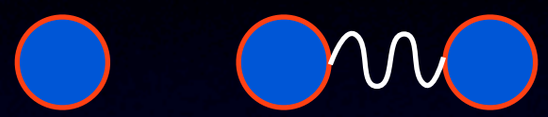


Require: degenerate through entire evolution

"Open loop" holonomic evolution

Adiabatic Teleportation

Initial Code



$|\psi\rangle \otimes |\Phi_+\rangle$

Final Code



$|\Phi_+\rangle \otimes |\psi'\rangle$

Initial Hamiltonian $\Delta > 0$ Final Hamiltonian

$$H_i = -\Delta(X_2X_3 + Z_2Z_3) \qquad H_f = -\Delta(X_1X_2 + Z_1Z_2)$$

$$H_i = -\Delta(S_1 + S_2) \qquad H_i = -\Delta(S_1' + S_2')$$

Evolution: $H(t) = \left(1 - \frac{t}{T}\right) H_i + \frac{t}{T} H_f$

Adiabatic Teleportation

$$\begin{aligned} \bar{X}_1 &= X_1 X_2 & \bar{Z}_1 &= Z_2 Z_3 \\ \bar{X}_2 &= X_2 X_3 & \bar{Z}_2 &= Z_1 Z_2 \\ \bar{X}_3 &= X_1 X_2 X_3 & \bar{Z}_3 &= Z_1 Z_2 Z_3 \end{aligned}$$

Initial Hamiltonian

$$H_i = -\Delta(\bar{Z}_1 + \bar{X}_2)$$

$$X_1 = \bar{X}_2 \bar{X}_3$$

$$Z_1 = \bar{Z}_1 \bar{Z}_3$$

Final Hamiltonian

$$H_f = -\Delta(\bar{X}_1 + \bar{Z}_2)$$

$$X_3 = \bar{X}_1 \bar{X}_3$$

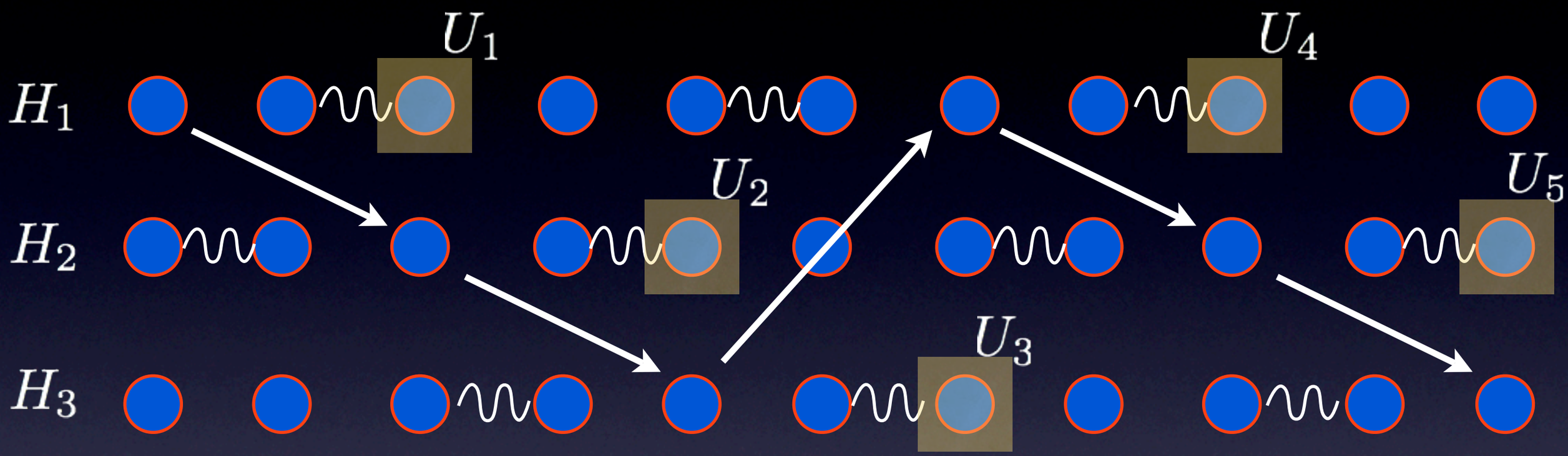
$$Z_3 = \bar{Z}_2 \bar{Z}_3$$

$$|\psi\rangle \otimes |\Phi_+\rangle$$



$$|\Phi_+\rangle \otimes |\psi\rangle$$

AQC Architectures



Sequence of Hamiltonians: $H_1, H_2, H_3, H_1, H_2, H_3, \ldots$

Sequence of single qubit gates applied: $U_1, U_2, U_3, U_4, U_5, \ldots$

Generalizing to more than 1 qubit:

Universal QC by cycling between only 3 Hamiltonians